

Jason Tang

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Education

University of Toronto

Sept. 2023 – May 2028

BASc, Computer Engineering | GPA: 3.66/4.00

- **Awards:** Edward S Rogers Sr. Scholarship, Faculty of Applied Science and Engineering Scholarship
- **Activities:** Tri-Campus Volleyball Captain; Badminton Team; Ultimate Frisbee Team Captain
- **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Operating Systems, Digital Systems, Computer Hardware, Computer Networking

Experience

AMD

May 2025 – Aug. 2026

Software Engineer Intern

Toronto, ON

- Developed and scaled an internal regression-management platform used by **2,000+ engineers**, supporting **3.1M tests** per month across a Vue, TypeScript, Node.js/Express, Ruby, and PostgreSQL stack
- Improved reliability and recovery for **239K jobs per month** by building IBM LSF lifecycle services that reconciled scheduler and application state across timeouts, failures, and abnormal terminations
- Enabled **100+ engineers** to execute **10K+ monthly** AI-assisted operations by developing an authenticated **TypeScript MCP** integration for failure investigation, coverage analysis, and controlled regression workflows
- Reduced initial regression-failure triage time from **15–30 minutes to approximately 30 seconds** by building an AI debugging agent that correlated logs, scheduler events, and regression metadata to identify likely failure causes
- Saved **20+ engineering hours per week** and shortened release cycles by **30%** by automating deployment workflows with object-oriented Python, Ruby, and Jenkins tooling
- Architected version-control abstractions and validation workflows for a **113-repository** Perforce-to-Git migration, securing full traceability for **1.4M+** historical regression runs

U+ Education

June 2023 – Sept. 2023

Web Developer

Ottawa, ON

- Designed and launched two responsive WordPress websites serving **100+ customers**, creating reusable layouts and streamlined content-management workflows
- Improved page-load performance by **30%** and increased organic traffic by **50%+** through performance optimization, technical SEO, and analytics-driven content updates

Digitera Interactive

June 2020 – Sept. 2020

Full-stack Developer Intern

Ottawa, ON

- Developed an end-to-end event-planning application prototype supporting event creation, scheduling, attendee registration, and account management using **HTML, CSS, JavaScript, and MySQL**
- Designed the **MySQL** database structure and implemented form validation, filtering, and error handling to support reliable CRUD operations

Projects

Mini ML Framework | Rust, CUDA, TypeScript, PyTorch, TensorFlow

Aug. 2026

- Built a custom ML framework in **Rust** with multidimensional tensors, reverse-mode automatic differentiation, CPU/CUDA backends, and **TypeScript** bindings for training and inference
- Validated tensor operations, gradients, and optimizer behavior against **PyTorch and TensorFlow** through cross-framework correctness tests
- Trained a 235K-parameter MLP on MNIST to **97.4%** test accuracy using cross-entropy loss and the Adam optimizer
- Developed custom **CUDA** kernels for elementwise operations and reductions, reducing training time from **4.1s to 1.3s** per epoch, a **3.2×** speedup over the CPU backend

FPGA 2048 | Verilog, VHDL, Quartus, ModelSim

Sept. 2024

- Built an **FPGA** implementation of 2048 in Verilog with gameplay rendered to a 160 × 120 VGA display at 60 Hz on a DE1-SoC board through a **custom VGA** adapter
- Implemented tile movement, merging, and scoring with a modular datapath and FSM controller, verifying the design through **ModelSim** testbenches, Tcl simulations, and physical I/O

Technical Skills

Languages: Python, TypeScript, JavaScript, Rust, C++, C, Ruby, SQL, SystemVerilog, Verilog, VHDL, Perl, HTML/CSS

Frameworks: Vue.js, React, Node.js, Express.js, UVM, Vite, PostgreSQL, PyTorch, TensorFlow

Technologies: YAML, Linux, Bash, Docker, Jenkins, Nginx, RabbitMQ, Git, Perforce, REST APIs, MCP, CUDA, AWS